Containerization_and_DevOps

📘 Experiment 9: Ansible

🧩 Problem Statement

Managing infrastructure manually across multiple servers leads to configuration drift, inconsistent environments, and repetitive tasks. Scaling becomes difficult with manual SSH-based administration.

🚀 What is Ansible?

Ansible is an open-source automation tool used for configuration management, application deployment, and orchestration.

Features:

- Agentless (uses SSH)
- YAML-based playbooks
- Idempotent (safe to run multiple times)
- Push-based architecture

⚙️ Key Concepts

Component Description Control Node Machine with Ansible installed Managed Nodes Target servers Inventory List of servers Playbooks YAML automation files Tasks Individual actions Modules Built-in functionalities Roles Reusable scripts

🔄 How Ansible Works

- Control node connects to managed nodes via SSH
- Executes modules through tasks
- Tasks grouped into playbooks

- Inventory defines servers

✅ Benefits

- Free & open source
- Easy to use
- No agents required
- Scalable
- Strong community support

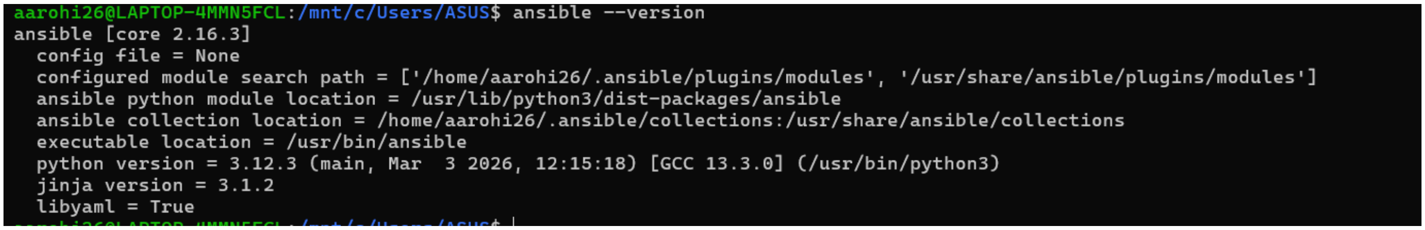
🧪 Part A: Installation

Install via pip

```
pip install ansible
```

```
C:\Users\ASUS>pip install ansible
Collecting ansible
  Downloading ansible-13.5.0-py3-none-any.whl.metadata (8.1 kB)
Collecting ansible-core~=2.20.4 (from ansible)
  Downloading ansible_core-2.20.4-py3-none-any.whl.metadata (7.7 kB)
Collecting jinja2>=3.1.0 (from ansible-core~=2.20.4->ansible)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting PyYAML>=5.1 (from ansible-core~=2.20.4->ansible)
  Downloading pyyaml-6.0.3-cp312-cp312-win_amd64.whl.metadata (2.4 kB)
Collecting cryptography (from ansible-core~=2.20.4->ansible)
  Downloading cryptography-46.0.7-cp311-abi3-win_amd64.whl.metadata (5.7 kB)
Requirement already satisfied: packaging in c:\users\asus\appdata\local\programs\python\python312\lib\site-packages (from ans
.1)
Collecting resolvelib<2.0.0,>=0.8.0 (from ansible-core~=2.20.4->ansible)
  Downloading resolvelib-1.2.1-py3-none-any.whl.metadata (3.7 kB)
Collecting MarkupSafe>=2.0 (from jinja2>=3.1.0->ansible-core~=2.20.4->ansible)
  Downloading markupsafe-3.0.3-cp312-cp312-win_amd64.whl.metadata (2.8 kB)
Collecting cffi>=2.0.0 (from cryptography->ansible-core~=2.20.4->ansible)
  Downloading cffi-2.0.0-cp312-cp312-win_amd64.whl.metadata (2.6 kB)
Collecting pycparser (from cffi>=2.0.0->cryptography->ansible-core~=2.20.4->ansible)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Downloading ansible-13.5.0-py3-none-any.whl (56.1 MB)
   ---------------------------------------- 56.1/56.1 MB 493.0 kB/s eta 0:00:00
Downloading ansible_core-2.20.4-py3-none-any.whl (2.4 MB)
   ---------------------------------------- 2.4/2.4 MB 604.8 kB/s eta 0:00:00
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloading pyyaml-6.0.3-cp312-cp312-win_amd64.whl (154 kB)
Downloading resolvelib-1.2.1-py3-none-any.whl (18 kB)
Downloading cryptography-46.0.7-cp311-abi3-win_amd64.whl (3.5 MB)
   ---------------------------------------- 3.5/3.5 MB 773.3 kB/s eta 0:00:00
Downloading cffi-2.0.0-cp312-cp312-win_amd64.whl (183 kB)
Downloading markupsafe-3.0.3-cp312-cp312-win_amd64.whl (15 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Installing collected packages: resolvelib, PyYAML, pycparser, MarkupSafe, jinja2, cffi, cryptography, ansible-core, ansible

[notice] A new release of pip is available: 25.0.1 -> 26.0.1
```

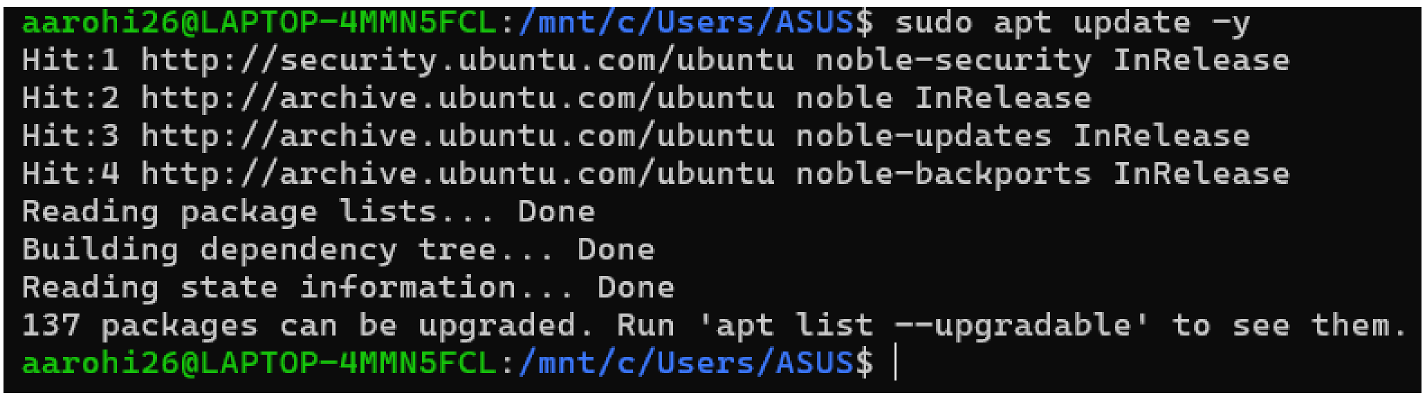
```
ansible --version
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/aarohi26/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/aarohi26/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar  3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

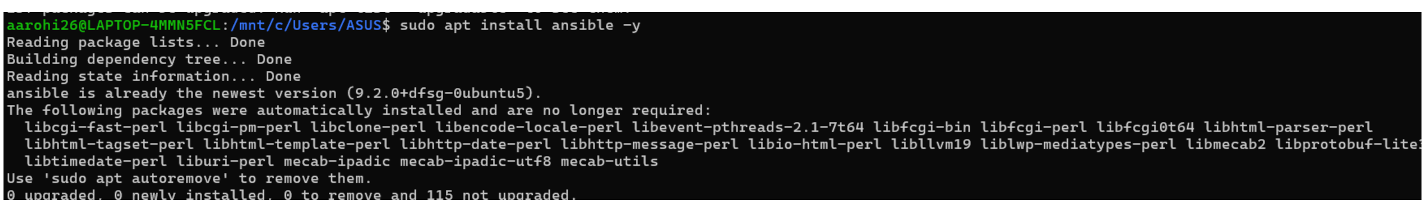
Install via apt

```
sudo apt update -y
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ sudo apt update -y
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
137 packages can be upgraded. Run 'apt list --upgradable' to see them.
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

```
sudo apt install ansible -y
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (9.2.0+dfsg-0ubuntu5).
The following packages were automatically installed and are no longer required:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl libllvm19 liblwp-mediatypes-perl libmecab2 libprotobuf-lite
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 115 not upgraded.
```

Test Installation

```
ansible localhost -m ping
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ ansible localhost -m ping
[WARNING]: No inventory was parsed, only implicit localhost is available
localhost | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

Expected Output: “ping”: “pong”

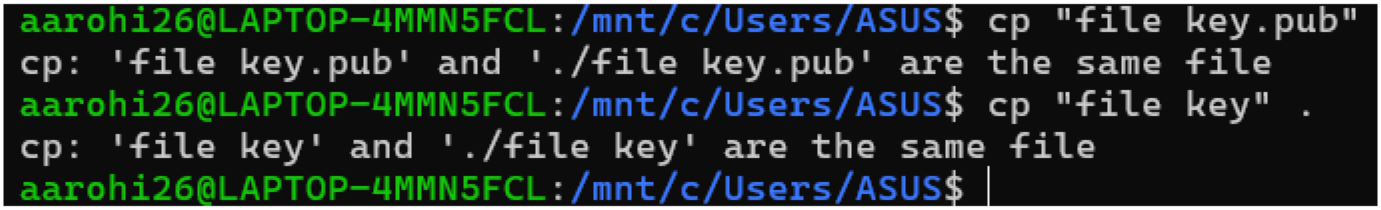
🐳 Docker + SSH Setup

Generate SSH Key

```
ssh-keygen -t rsa -b 4096
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/aarohi26/.ssh/id_rsa): file key
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in file key
Your public key has been saved in file key.pub
The key fingerprint is:
SHA256:fUQeoT1rk0BFFzpBYxkfajRR9F+Uz2sEIIQeaZ0D6uM aarohi26@LAPTOP-4MMN5FCL
The key's randomart image is:
+---[RSA 4096]----+
|       .*oo=^B*.o|
|      .= =.B.@ = |
|     .o . + X o.+|
|    .  . . + = .=|
|     o  S . * . o|
|    . .    o . o |
|     E         . |
|                 |
|                 |
+----[SHA256]-----+
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS$
```

```
cp ~/.ssh/id_rsa.pub .
cp ~/.ssh/id_rsa .
```



Create Dockerfile

```
FROM ubuntu

RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/sshd

RUN mkdir -p /run/sshd && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config &
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config &&
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

RUN mkdir -p /root/.ssh && chmod 700 /root/.ssh

COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

RUN chmod 600 /root/.ssh/id_rsa && chmod 644 /root/.ssh/authorized_keys

RUN sed -i 's@session\\s*required\\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc

EXPOSE 22

CMD ["/usr/sbin/sshd", "-D"]
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ nano Dockerfile
```

```
aarohi26@LAPTOP-4MMN5F(
 GNU nano 7.2                                                                  Dockerfile
FROM ubuntu


RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/sshd

# Configure SSH
RUN mkdir -p /run/sshd && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config


# Create .ssh directory and set proper permissions
RUN mkdir -p /root/.ssh && \
    chmod 700 /root/.ssh

# Copy SSH keys (note: this is not secure for production!)
COPY file_key.pub /root/.ssh/authorized_keys

# Set proper permissions for keys
RUN chmod 600 /root/.ssh/id_rsa && \
    chmod 644 /root/.ssh/authorized_keys

# Fix for SSH login
RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/sshd

# Expose SSH port
EXPOSE 22

# Start SSH service when container starts
CMD ["/usr/sbin/sshd", "-D"]
```

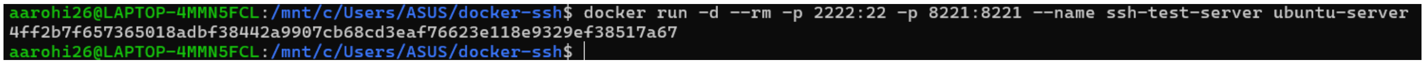
Build Image

```
docker build -t ubuntu-server .
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ docker build -t ubuntu-server .
[+] Building 29.5s (14/14) FINISHED                                                            docker:default
 => [internal] load build definition from Dockerfile                                                      0.0s
 => => transferring dockerfile: 1.03kB                                                                    0.0s
 => [internal] load metadata for docker.io/library/ubuntu:latest                                          0.1s
 => [internal] load .dockerignore                                                                         0.0s
 => => transferring context: 2B                                                                           0.0s
 => [1/9] FROM docker.io/library/ubuntu:latest@sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c  0.0s
 => => resolve docker.io/library/ubuntu:latest@sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c  0.0s
 => [internal] load build context                                                                         0.0s
 => => transferring context: 34B                                                                          0.0s
 => CACHED [2/9] RUN apt update -y                                                                        0.0s
 => CACHED [3/9] RUN apt install -y python3 python3-pip openssh-server                                    0.0s
 => CACHED [4/9] RUN mkdir -p /var/run/sshd                                                               0.0s
 => CACHED [5/9] RUN mkdir -p /run/sshd &&     echo 'root:password' | chpasswd &&     sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/s  0.0s
 => CACHED [6/9] RUN mkdir -p /root/.ssh &&     chmod 700 /root/.ssh                                      0.0s
 => CACHED [7/9] COPY file_key.pub /root/.ssh/authorized_keys                                             0.0s
 => [8/9] RUN chmod 644 /root/.ssh/authorized_keys                                                        0.2s
 => [9/9] RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/sshd  0.3s
 => exporting to image                                                                                   28.8s
 => => exporting layers                                                                                  22.9s
 => => exporting manifest sha256:5d20459d4032af3746e653695b538bd9b98c407e2b08a329943993d326f9279a          0.0s
 => => exporting config sha256:8ede1ad7c335d196a80ff5f2e2b3823797e273d9b8ab72296cbfc423fe0e6830            0.0s
 => => exporting attestation manifest sha256:4443ea098df5586f48abb9eabad37d96a0b644b0a98b278a4f57c3e5c27ad031  0.0s
 => => exporting manifest list sha256:7e7c249bd2ed06d30d59a949f95c9840257ca506eb9272066a458026306c8346     0.0s
 => => naming to docker.io/library/ubuntu-server:latest                                                   0.0s
 => => unpacking to docker.io/library/ubuntu-server:latest                                                5.8s
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ |
```

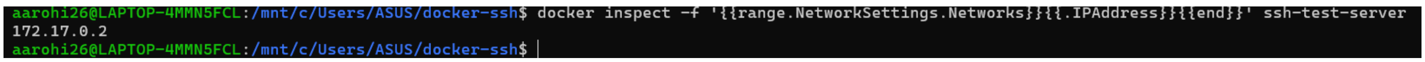
Run Container

```
docker run -d -p 2222:22 --name ssh-test-server ubuntu-server
```



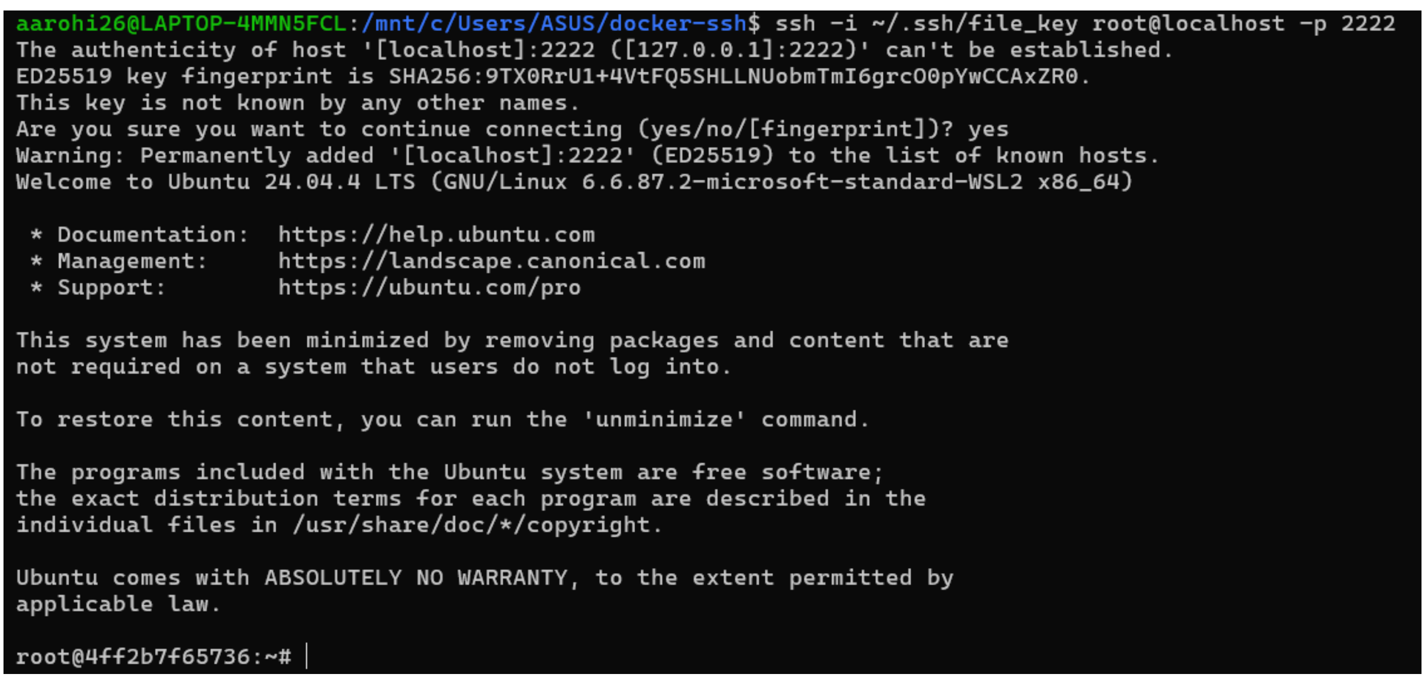
Get Container IP

```
docker inspect -f '' ssh-test-server
```



Test SSH

```
Password:
ssh root@localhost -p 2222
```




```
Key-based:
ssh -i ~/.ssh/id_rsa root@localhost -p 2222
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ ssh -i ~/.ssh/file_key root@localhost -p 2222
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Sat Apr 11 04:14:21 2026 from 172.17.0.1
root@4ff2b7f65736:~#
```

⚡ Ansible with Docker

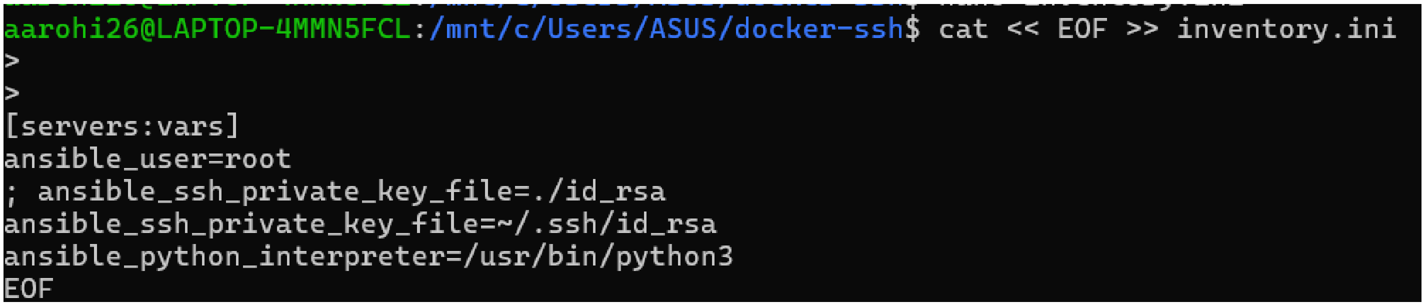
Run Multiple Servers

```
for i in {1..4}; do
  docker run -d -p 220${i}:22 --name server${i} ubuntu-server
done
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ for i in {1..4}; do
  echo -e "\n Creating server${i}\n"
  docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
  echo -e "IP of server${i} is $(docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i})"
done
0

 Creating server1

442a1acf8cf89788d2fa5c4e0162ed61d59c30b4ffe9ad6329b55c7e144a5e7e
IP of server1 is 172.17.0.3

 Creating server2

fd3ee61ae0f91446786695d73cd5d41fcdfb349cd3420c2ffbf5985ce8884239
IP of server2 is 172.17.0.4

 Creating server3

2a01e08afd7f32274f59a3a921746c88eb7214fc0e2264e6886a15c7b2e94c55
IP of server3 is 172.17.0.5

 Creating server4

6935cd7704c8d9ac24f7779b86a0cb67117434e9eb3b8ac5d7102437234655d4
IP of server4 is 172.17.0.6
0: command not found
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$
```

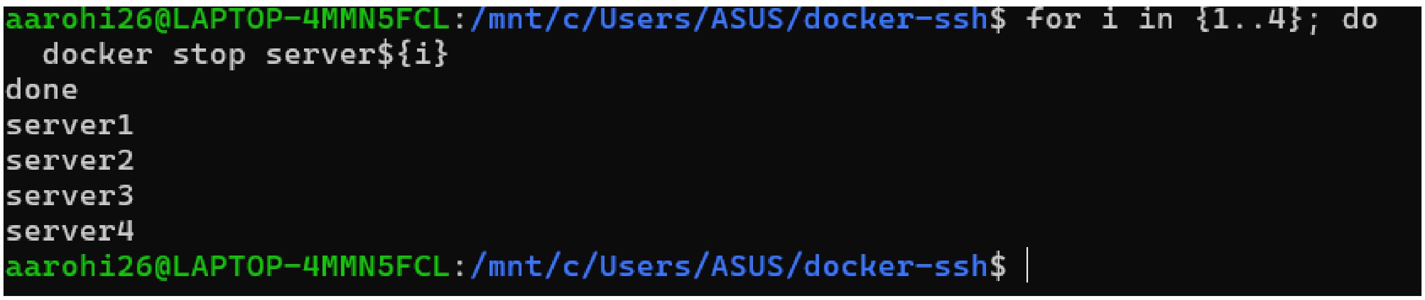
Create Inventory

```
echo "[servers]" > inventory.ini
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ cat << EOF >> inventory.ini
>
>
[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

```
for i in {1..4}; do
  docker inspect -f '' server${i} >> inventory.ini
done
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ for i in {1..4}; do
  docker stop server${i}
done
server1
server2
server3
server4
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$
```

```
cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ cat inventory.ini
[servers]
172.17.0.3
172.17.0.4
172.17.0.5
172.17.0.6
[servers]
server1 ansible_host=127.0.0.1 ansible_port=2201
server2 ansible_host=127.0.0.1 ansible_port=2202



[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$
```

Test Connectivity

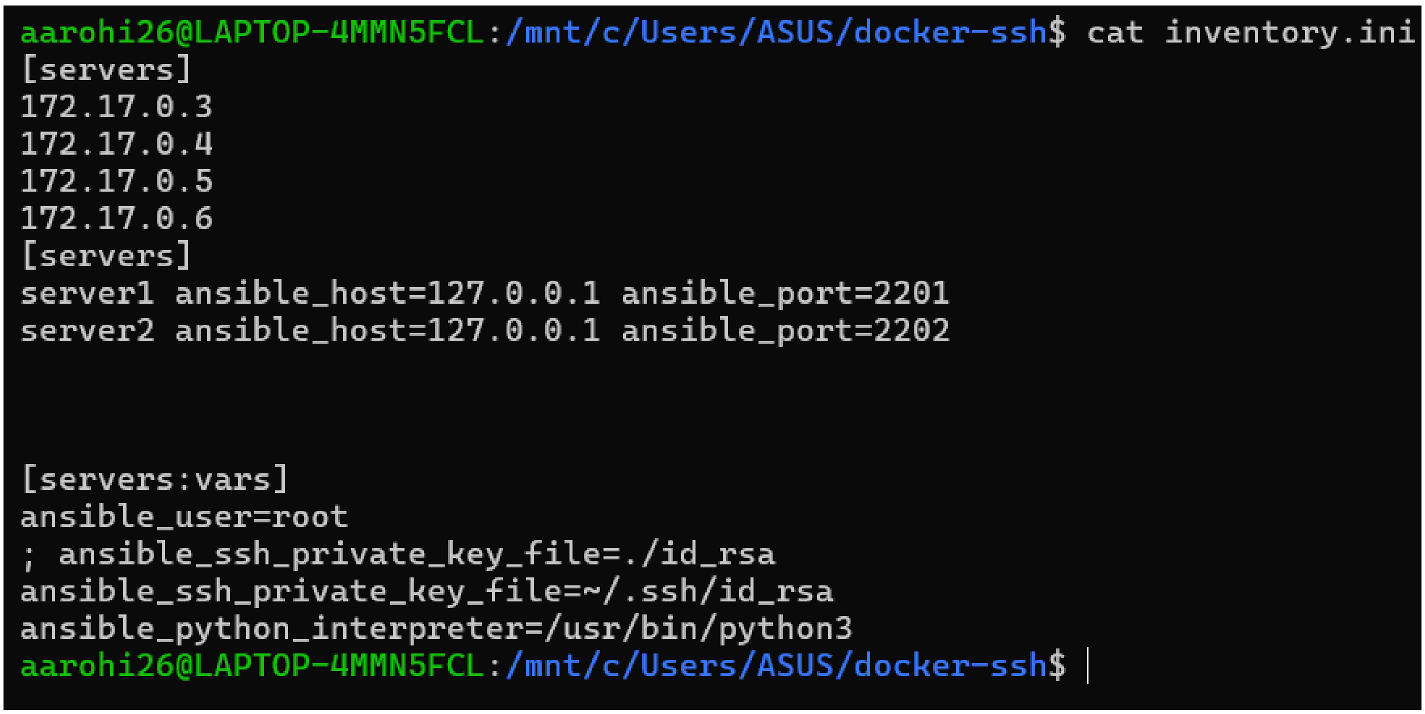
```
ansible all -i inventory.ini -m ping
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ cat << EOF >> inventory.ini
>
>
[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
```

Create Playbook (playbook1.yml)

```
---
- name: Update and configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update apt packages
      apt:
```

```
        update_cache: yes
        upgrade: dist

    - name: Install packages
      apt:
        name: ["vim", "htop", "wget"]
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on "
```



```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ cat inventory.ini
[servers]
172.17.0.3
172.17.0.4
172.17.0.5
172.17.0.6
[servers]
server1 ansible_host=127.0.0.1 ansible_port=2201
server2 ansible_host=127.0.0.1 ansible_port=2202



[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$
```

Run Playbook

```
ansible-playbook -i inventory.ini playbook1.yml
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ ansible all -i inventory.ini -m ping
server4 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$
```

Verify Output

```
ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
```

```
aarohi26@LAPTOP-4MMN5FCL:/mnt/c/Users/ASUS/docker-ssh$ nano update.yml
```

Cleanup

```
for i in {1..4}; do docker rm -f server${i}; done
```

aarohi26@LAPTOP-4MMN5F(Windows PowerShell

```
GNU nano 7.2
---
- name: Update and configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update apt packages
      apt:
        update_cache: yes
        upgrade: dist

    - name: Install required packages
      apt:
        name:
          - vim
          - htop
          - wget
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on {{ inventory_hostname }}"
```

📊 Need for Ansible

- Scalability
- Consistency
- Efficiency
- Idempotency
- Infrastructure as Code

⭐ Key Features

- Agentless
- Idempotent
- YAML-based

- Large module library
- Push-based

🎯 Conclusion

Ansible simplifies server management by automating repetitive tasks, ensuring consistency, and enabling scalable infrastructure management.